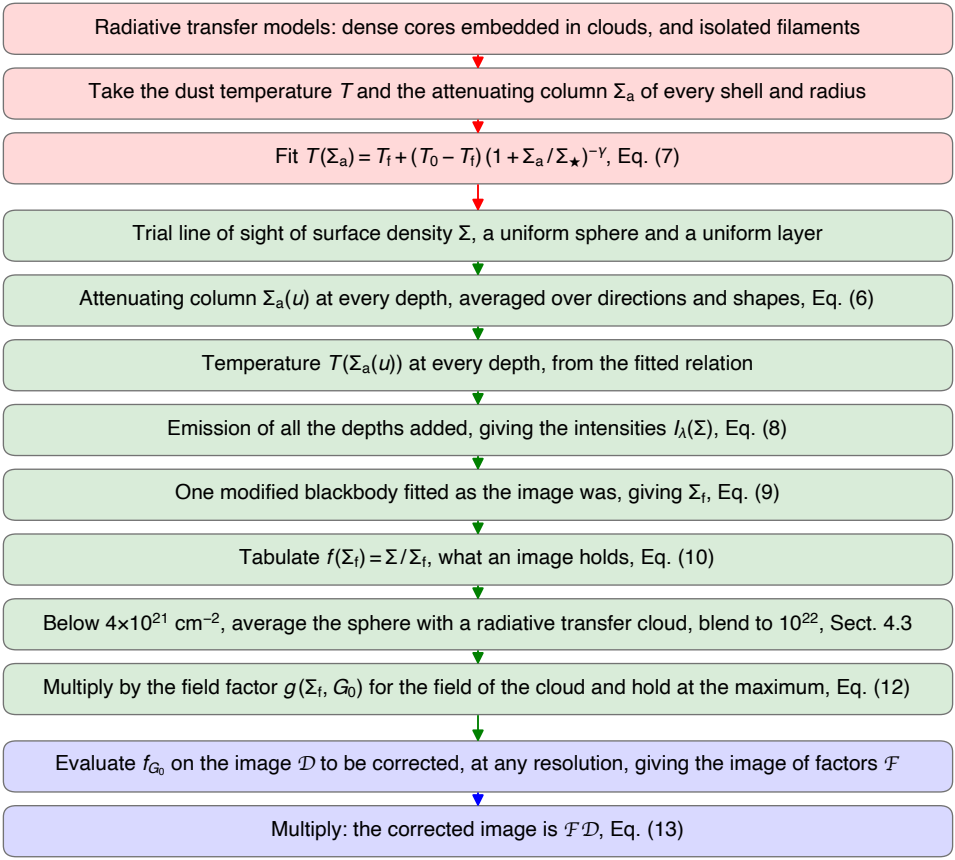
Radiative transfer models: dense cores embedded in clouds, and isolated filaments
Take the dust temperature T and the attenuating column Σ_a of every shell and radius
Fit $T(\Sigma_a) = T_f + (T_0 - T_f)(1 + \Sigma_a/\Sigma_\star)^{-\gamma}$, Eq. (7)
Trial line of sight of surface density Σ, a uniform sphere and a uniform layer
Attenuating column $\Sigma_a(u)$ at every depth, averaged over directions and shapes, Eq. (6)
Temperature $T(\Sigma_a(u))$ at every depth, from the fitted relation
Emission of all the depths added, giving the intensities $I_\lambda(\Sigma)$, Eq. (8)
One modified blackbody fitted as the image was, giving Σ_f, Eq. (9)
Tabulate $f(\Sigma_f) = \Sigma/\Sigma_f$, what an image holds, Eq. (10)
Below 4×10^{21} cm^{-2}, average the sphere with a radiative transfer cloud, blend to 10^{22}, Sect. 4.3
Multiply by the field factor $g(\Sigma_f, G_0)$ for the field of the cloud and hold at the maximum, Eq. (12)
Evaluate f_{G_0} on the image $\mathcal{D}$ to be corrected, at any resolution, giving the image of factors $\mathcal{F}$
Multiply: the corrected image is $\mathcal{F}\mathcal{D}$, Eq. (13)